

ACQUADRO PATRIZIO

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Robot Learning & Manipulation: multimodal perception, simulation, and real-robot systems

Skills

Core: Python, PyTorch, Isaac Sim, Isaac Lab, MuJoCo, HF Transformers, C, Java, NumPy, Pandas, SQL, R.

Tools & Software: Git, Linux, Docker, Apptainer, Slurm, DeepSpeed, Weights & Biases, pytest, LangChain, LaTeX.

Languages: Italian (native), English (fluent; IELTS 7.5), French (basic; 2 years, 6 hours/week).

Work & Research Experience

R&D Assistant – Robot Learning, Purdue University

Jun 2026 – Present

- Build simulation and control infrastructure for humanoid manipulation, and perception systems supporting ONR research.
- Mentor an undergraduate researcher on Ego2Grip, guiding technical development.

Visiting Student Researcher – Purdue University (E-Lab)

Jan 2026 – May 2026

- Integrated a decoder-only transformer policy on the IHMC Alex humanoid for real-time validation.
- Owned the end-to-end experimentation pipeline, optimizing for reproducibility, tracking, and rapid ablations.

Generative AI Researcher – Politecnico di Milano

Feb 2025 – Sep 2025

- Improved grounding for repo-level codegen via retrieval/reranking ablations (BM25, CodeBERT, RRF, multi-hop).
- Curated a 7.6k-snippet KB (PySCF, SEED-Emulator), standardizing chunking and filtering for coverage.
- Optimized iteration speed and GPU cost with 4-bit NF4, caching, and resume/OOM-safe runs for large sweeps.
- STM32N6 project: Ported an INT8 transformer to a CNN-oriented NPU; validated with ST engineers.

Applied ML Researcher – University of Milano-Bicocca

Mar 2024 – Jun 2024

- Designed LangChain RAG with pairwise re-ranking to improve clinical-trial ranking and reduce noise.
- Built a UMLS-based KB for grounding/normalization and benchmarked OSS vs GPT API on accuracy/latency/cost.

AI Research Intern (Quantum ML) – Exprivia S.p.A.

Apr 2024 – Jun 2024

- Evaluated QML feasibility for Heston modeling by benchmarking PINN/WGAN baselines against a qWGAN in Qiskit.

Selected Projects & Entrepreneurship

AlexDoor-XAS — Developing an IHMC Alex door-opening benchmark to compare four action representations with ACT/diffusion policies and evaluate generalization to unseen doors.

Isaac Audio Sensors — Built an open-source Isaac Sim/Lab SDK for microphone-array simulation and sound propagation, with spatial observations, recording/replay, and robot-learning datasets.

SquadBot-AV — Developing audio-visual perception that uses sound cues to guide visual search and confirm sound sources.

Ego2Grip — Extracting 3D hand trajectories from first-person video toward robot-compatible manipulation demonstrations.

VLA-LEGO — Built a contact-rich MuJoCo benchmark for bimanual assembly to assess robustness and failure modes.

BiellaInFesta, Instagram Social Page – Founder

Jul 2022 – Present

Grew a local events page to 8.9k followers and 550k/mo views, and leading a 4+ person team.

Adunet WebApp – Founder

Apr 2025 – May 2025

Shipped a Next.js festival map app, and leading a 10-person promo team serving 8–10k daily visitors.

CubeWar Minecraft Server – Co-founder

May 2018 – Jul 2019 (sold)

Scaled a Java Minecraft server to 70–90 daily players (300 peak), and running ops in a 4-person team.

Education

Purdue University – Visiting Exchange Student

Aug 2025 – May 2026

Selected on merit (1 of 2) from a 674-student first-year MSc CS cohort (PoliMi).

Politecnico di Milano – M.S. Computer Science & Engineering (AI)

Sep 2024 – Present

GPA 28.2/30 (Top 8% - Polimi Open Data); Honors (30L/30): LLMs (PhD course), SWE II.

University of Milano-Bicocca – M.S. AI for Science and Technology

Sep 2024 – Present

Top grades (30/30): Advanced AI/ML/DL, Embedded Systems.

University of Pavia – B.S. Artificial Intelligence

Sep 2021 – Oct 2024

Degree grade: 110/110; First inter-university B.S. in AI in Italy.

University Scholarships: Merit-based, income-qualified (awarded 2023–2024–2025, €5,500/year)

Additional

Interests: Swimming (3 years competitive), rock climbing (regional competitions), philosophy, and fantasy literature.

Community: JCI Biella: led communications for 4 events; Italian Red Cross: certified first-aid volunteer.